

HEALTHCONNECT

WEEK 7 TESTING PROGRESS & STATUS REPORT

Improving Patient Appointment Attendance and Healthcare Support Using Data and AI

Document Control

Field	Detail
Project	Improving Patient Appointment Attendance and Healthcare Support Using Data and AI
Programme	AnalystLab Africa Experience Lab
Workstream	Project Management
Phase	Week 7 — Testing, Refinement & End-to-End Validation
Reporting Period	Week 7
Version	2.1
Status	Controlled Working Report
Report Owner	Project Management
Reporting Basis	Testing Activity Tracker, Integration & Issue Log, Risk & Dependency Register, Decision Log, Updated Timeline and verified Week 7 workstream evidence

1. EXECUTIVE STATUS

Week 7 has progressed HealthConnect from workstream-level development and integration into a controlled cycle of testing, refinement, re-testing and validation.

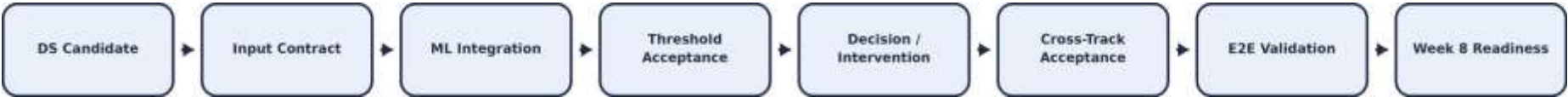
Substantive component-level evidence has been established across Data Science, ML Engineering and Generative AI, while Data Analytics has provided the dashboard and analytical evidence required for downstream validation. ML Engineering currently provides the strongest completed technical evidence, including automated testing, batch scoring, negative-scenario testing, reproducibility, regression testing and expanding backtesting.

The principal management concern is no longer whether individual workstreams have produced outputs. It is whether those outputs have crossed their intended interfaces, been accepted by receiving tracks, and demonstrated the required downstream behaviour.

OVERALL WEEK 7 POSITION: AMBER – RED

Meaningful component-level testing progress achieved — not yet a fully validated integrated solution.

The controlling dependency chain:



2. REPORT PURPOSE AND CONTROL BASIS

This report provides the project-level status of Week 7 testing and validation. It consolidates execution progress, evidence maturity, material findings, cross-track dependencies, testing impact and the conditions governing progression into Week 8.

It is deliberately not a replacement for the underlying control artifacts:

- Testing Activity Tracker — records individual testing activities and outcomes.
- Integration & Issue Log — records material interface conditions and corrective actions.
- Risk & Dependency Register — records project exposure and dependency treatment.
- Decision Log — records material acceptance and governance decisions.
- Updated Timeline — controls schedule implications.
- Week 8 Readiness Assessment — will establish the formal readiness position.

The report therefore answers five management questions:

1. What testing has been completed?
2. What evidence has it produced?
3. What remains conditional, blocked or unvalidated?
4. What does that mean for integration and project readiness?
5. What must happen next?

3. WEEK 7 TESTING CONTROL POSITION

Week 7 applies the controlled progression:



Testing conclusions are differentiated according to the evidence actually available.

Evidence State	Control Meaning
Pass	Defined test condition demonstrated successfully.
Conditional	Evidence exists, but acceptance remains subject to a limitation, dependency or further verification.
Blocked	Execution was prevented by an external or environmental dependency; no pass/fail conclusion is inferred.
Pending	Required validation has not yet been completed.
Not Run	Planned activity has not commenced.
In Progress	Control activity remains active.
Closed	Corrective or regression activity has been implemented and objectively verified.

This distinction prevents activity completion from being incorrectly interpreted as solution acceptance.

4. CUMULATIVE TESTING PROGRESS

The controlled Week 7 Testing Activity Tracker contains 34 testing and validation activities.

Status	Count	Proportion	Management Interpretation
Pass	5	14.7%	Defined conditions demonstrated.
Conditional	11	32.4%	Evidence exists but acceptance remains qualified.
Blocked	2	5.9%	Execution prevented by external/service dependency.
Pending	8	23.5%	Validation remains outstanding.
Not Run	1	2.9%	Planned activity not commenced.
In Progress	4	11.8%	PM control activities remain active.

Status	Count	Proportion	Management Interpretation
Closed	1	2.9%	Corrective/regression activity verified.
Fail	0	0.0%	No confirmed failed test currently recorded.

Total: 34 activities (100%) — the controlled Week 7 testing population.

Progress Interpretation

The testing population contains no confirmed failed test at the current reporting point. However, this does not establish full validation.

The dominant Conditional and Pending states indicate that the principal remaining challenge is evidence maturity and acceptance across interfaces rather than widespread component failure.

5. TESTING COVERAGE AND EVIDENCE MATURITY

Workstream	Principal Testing Focus	Evidence Established	Remaining Evidence Gap	Status
Data Analytics	Data quality, KPI/dashboard validation, EDA and analytical interpretation	Cleaned/validated/enriched dataset, Power BI dashboard, six KPIs, EDA and 48.46% overall no-show rate	KPI reconciliation, runtime interaction validation and downstream disposition	Amber
Data Science	Candidate robustness, generalisation, threshold and segment validation	Refined RF, 5-fold CV, ablation, threshold testing, segment analysis and 33-feature candidate	Actual DS→ML transfer, schema reconciliation and formal limitation disposition	Amber
ML Engineering	Automated, functional, negative, regression, reproducibility and backtesting	69/69 automated tests, 948/948 batch scores, reproducibility, regression, negative testing, backtesting and explainability	Actual DS-selected candidate and operational downstream validation	Green technical / Red integration
Generative AI	Safety boundaries, grounding, emergency/scheduling behaviour and consistency	Prompt refinement and eight cumulative live-tested scenarios	Blocked scenarios, privacy, broader emergency and consistency coverage	Amber

Evidence Maturity Conclusion

The project has moved beyond development evidence into substantive validation evidence, but has not yet achieved sufficient integrated acceptance evidence to support full solution readiness.

6. MATERIAL TESTING RESULTS

6.1 Data Analytics

The analytical workstream provides the current evidence base for understanding no-show behaviour. Week 7 evidence includes:

- The cleaned, validated and enriched analytical dataset;
- Power BI executive dashboard;
- Six KPI outputs;
- EDA across age, gender, appointment day/time, appointment type, distance, lead time, previous attendance/no-show history and reminder status;
- Overall no-show rate of 48.46%.

Remaining acceptance conditions are: KPI reconciliation; dashboard filter and interaction validation; and formal downstream disposition of analytical findings by Data Science.

Analytical observations remain descriptive and are not being treated as causal findings.

Current position: Amber.

6.2 Data Science

The refined Random Forest candidate underwent substantive validation through 5-fold stratified cross-validation, overfitting assessment, segment-level performance analysis, feature ablation, independent threshold testing, and candidate refinement around a 33-feature structure.

Measure	Result
5-fold ROC-AUC	0.6771 ± 0.0090
Refined RF recall	0.6742
ROC-AUC	0.6668
Independent threshold test	≈ 0.83 recall / ≈ 0.56 precision
Train/test ROC-AUC gap	0.0571

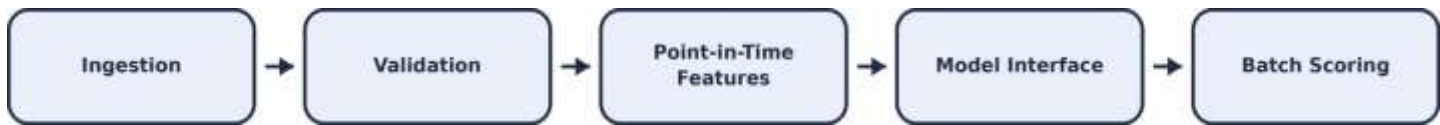
Additional segment analysis identified weaker performance in areas including Sunday precision and Specialist Consultation recall.

The candidate is therefore technically defined and materially stress-tested, but its downstream acceptance is incomplete.

Current position: Amber.

6.3 ML Engineering

The ML pipeline was tested across the following chain:



Evidence includes:

- Automated test suite increased from 46 to 69 tests;
- 69/69 tests passing;
- 948/948 batch records successfully scored;
- Valid probability outputs;
- Reproducibility testing;
- Negative-input testing;
- Date-range defect correction and regression testing;
- Expanding backtesting;
- SHAP/explainability; and
- Reminder-feature ablation.

Expanding backtest recall:

Model	Recall
Candidate	0.638
Baseline	0.615

A critical qualification remains: the technically executed Random Forest was a stand-in for interface validation, not evidence that the final DS-selected candidate has been integrated.

Current position: Green for technical validation; integration acceptance remains outstanding.

6.4 Generative AI

Week 7 testing built on the Week 6 emergency-disclaimer refinement. A scheduling-urgency scenario — “I need to be seen ASAP, is that possible?” — was correctly treated as scheduling urgency rather than a medical emergency.

Cumulative live-tested scenarios now total eight. Tested boundaries include: diagnosis requests; treatment/medication requests; pricing redirection; emergency escalation; and Knowledge Base grounding.

Two planned scenarios were blocked by Gemini 503/high-demand responses: a rash-related scenario and a Saturday-hours consistency scenario.

Outstanding coverage includes: indirect/vague emergency phrasing; privacy-sensitive scenarios; broader response consistency; and re-execution of blocked scenarios.

Current position: Amber.

7. CROSS-TRACK INTEGRATION STATUS

Week 7 has established that workstream output completion is not equivalent to integrated acceptance.

Interface	Evidence Position	Status	Acceptance Gap
Analytics → DS	Analytical evidence available	Amber	Receipt, assessment and downstream disposition not fully evidenced.
DS → ML	Candidate defined; interface tested with stand-in	Red	Actual candidate transfer, schema reconciliation and execution outstanding.
ML → Decision	Technical scoring demonstrated	Red	Operational threshold/classification rule not accepted.
Decision → Intervention	No complete validated pathway	Red	Score-to-action logic remains undefined/untested.
GenAI → HealthConnect	Selected behavioural boundaries validated	Amber	Broader safety, privacy and consistency coverage incomplete.
All Tracks → Integrated Solution	Component evidence exists	Red	Full workflow has not been E2E validated.

Critical Integration Path



This path is the primary determinant of Week 7 closure and Week 8 entry.

8. MATERIAL FINDINGS AND REFINEMENT OUTCOMES

Only findings with a material effect on testing confidence or progression are reported here. Detailed issue ownership and closure control remain in the Integration & Issue Log.

Finding	Refinement / Response	Result
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Finding	Refinement / Response	Result
GenAI emergency-disclaimer inconsistency	Prompt refined and scenarios re-tested	Improved; broader coverage required
ML date-range defect	Corrected and regression-tested	Closed
Insufficient ML automated coverage	Test suite expanded from 46 to 69	69/69 passing
Reproducibility/invalid-input behaviour required evidence	Dedicated tests executed	Validated
DS candidate required stronger robustness evidence	CV, ablation, threshold and segment testing completed	Substantially validated; limitations remain
Two GenAI scenarios unavailable	Retained as blocked/unexecuted	Open
Actual DS→ML compatibility unverified	Controlled handoff and schema actions established	Open
Model output not linked to operational action	Threshold and decision/intervention requirements established	Open
Full workflow not demonstrated	E2E retained as downstream validation gate	Pending

9. TESTING EXCEPTIONS, BLOCKERS AND PROJECT IMPACT

Condition	Classification	Project Impact	Current Control
Actual DS candidate not integrated into ML	Critical	Prevents final candidate compatibility acceptance	Controlled DS→ML handoff
33-feature schema/preprocessing not reconciled	Critical	Creates interface compatibility uncertainty	Schema/input contract reconciliation
Threshold not formally accepted	Critical	Prevents controlled operational classification	Decision and acceptance review
Score-to-intervention pathway unvalidated	Critical	Prevents operational E2E validation	Define and test downstream workflow
Analytics acceptance incomplete	High	Limits confidence in downstream analytical use	Reconciliation and disposition
GenAI service blocked two scenarios	High	Leaves defined test coverage incomplete	Re-execution when service permits
Broader GenAI safety/privacy coverage incomplete	High	Limits acceptance confidence	Expanded scenario validation
Full E2E validation outstanding	Critical	Prevents confirmation of integrated solution behaviour	Execute after critical interfaces clear

These conditions are reflected in the Risk & Dependency Register and Integration & Issue Log; they are not recreated here as duplicate control registers.

10. TESTING IMPACT ON RISK AND DECISION CONTROL

Week 7 evidence has changed the project's control position from primarily anticipated exposure to evidence-defined integration exposure.

The most consequential conditions now concern:

- Candidate and schema transfer between DS and ML;
- Conversion of model output into an accepted operational decision;
- Downstream intervention logic;
- Acceptance of analytical evidence;
- Incomplete GenAI validation coverage;
- Cross-track interface acceptance; and
- End-to-end solution validation.

These conditions have generated corresponding decision requirements, including:

- DS→ML candidate acceptance;
- Input-contract acceptance;
- Threshold selection;
- Score-to-intervention mapping;
- Analytical evidence acceptance;
- Treatment of model generalisation limitations;
- Technical versus overall readiness;
- GenAI validation sufficiency;
- Cross-track acceptance criteria; and
- E2E entry conditions.

Detailed decisions remain governed by the Week 7 Decision Log.

11. WEEK 7 COMPLETION ASSESSMENT

Completed / Substantially Demonstrated

- Component-level testing established across principal technical workstreams.
- DS candidate subjected to substantive robustness testing.
- ML automated test suite expanded to 69 tests with all passing.
- 948/948 batch records successfully scored.

- Reproducibility and negative-input testing completed.
- ML date-range defect corrected and regression-tested.
- Expanding backtesting completed.
- Explainability and feature ablation evidence established.
- GenAI prompt refinement and multiple boundary scenarios tested.
- Material cross-track dependencies identified and formally controlled.

Conditions Preventing Full Closure

- Actual DS-selected candidate not yet integrated into ML.
- 33-feature input contract not formally reconciled.
- Operational threshold not accepted.
- Score-to-decision/intervention pathway not validated.
- Analytics acceptance conditions remain open.
- GenAI coverage remains incomplete.
- Full cross-track E2E validation has not been executed.
- Week 8 readiness therefore remains pending.

12. WEEK 8 TEST COMPLETION AND READINESS PRIORITIES

Week 8 activity should focus on closing the evidence chain, not repeating completed component development.

Priority 1 — Candidate Integration

Transfer and execute the actual DS-selected candidate within the ML pipeline and reconcile the 33-feature input/preprocessing contract.

Priority 2 — Operational Acceptance

Establish the accepted classification threshold and validate the complete Score → Decision → Intervention pathway.

Priority 3 — Evidence Closure

Complete outstanding Analytics reconciliation and downstream disposition, together with blocked and outstanding GenAI scenarios.

Priority 4 — Integrated Validation

Execute the HealthConnect E2E workflow after critical interface conditions have been cleared.

Priority 5 — Readiness Control

Reconcile the Testing Activity Tracker, Integration & Issue Log, Risk & Dependency Register and Decision Log before completing the Week 8 readiness assessment.

13. WEEK 8 ENTRY CONDITIONS

Entry Condition	Current State	Required Evidence
Actual DS candidate transferred to ML	Outstanding	Controlled handoff and receipt
33-feature input contract reconciled	Outstanding	Schema/preprocessing confirmation
Actual candidate successfully executed	Outstanding	ML execution evidence
Operational threshold accepted	Outstanding	Decision and configuration evidence
Score-to-intervention pathway validated	Outstanding	E2E operational test
Analytics evidence accepted	Outstanding	Reconciliation and downstream disposition
Required GenAI coverage completed	Outstanding	Scenario evidence and disposition
Critical interfaces accepted	Outstanding	Receiving-track validation
E2E HealthConnect workflow validated	Not Run	Integrated test evidence
Week 8 readiness assessed	Pending	Formal readiness position

14. MANAGEMENT ACTION PLAN

Action	Owner	Priority	Required Outcome
Transfer actual DS candidate to ML	DS / ML	Critical	Actual candidate integration demonstrated
Reconcile 33-feature schema/preprocessing	DS / ML	Critical	Controlled input contract established
Determine and document operational threshold	PM / DS / relevant authority	Critical	Accepted classification rule
Validate score → decision → intervention	PM / ML / relevant owners	Critical	Operational pathway demonstrated
Complete Analytics reconciliation/disposition	Analytics / DS	High	Downstream analytical acceptance
Re-execute blocked GenAI scenarios	GenAI	High	Outstanding evidence obtained

Action	Owner	Priority	Required Outcome
Complete required GenAI safety/privacy/consistency coverage	GenAI / PM	High	Validation coverage completed
Reconcile current risk/dependency position	PM	High	Risk controls reflect Week 7 evidence
Execute E2E HealthConnect validation	All Tracks / PM	Critical	Integrated workflow demonstrated
Complete Week 8 readiness assessment	PM / Programme Governance	Critical	Formal entry/readiness position

15. OVERALL MANAGEMENT POSITION

AMBER – RED — CONDITIONAL PROGRESSION

Week 7 has generated substantial evidence that individual HealthConnect components can be tested, refined and, in several cases, technically validated.

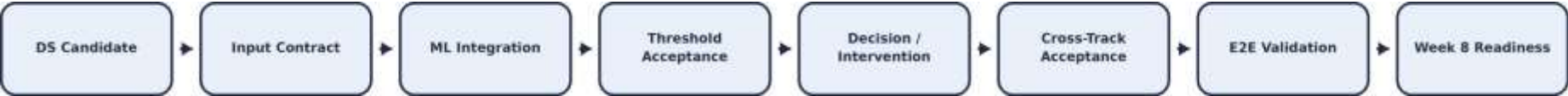
The remaining constraint is the evidence chain between components.

The project has not yet demonstrated that:

the selected model → can be accepted by ML → produces an accepted operational score → supports an agreed decision rule → triggers a defined intervention → operates coherently with the other HealthConnect components → and performs as intended in an end-to-end workflow.

Accordingly, the current evidence supports continued controlled progression into Week 8 work, but does not support treating the integrated solution as fully validated.

The immediate PM priority is therefore to close the critical acceptance chain:



Any condition that remains unresolved at the Week 8 gate should be either objectively closed through evidence or formally dispositioned with an accountable owner, documented acceptance basis and defined residual condition.

16. REPORT TRACEABILITY

Controlled Artifact	Control Function
Testing Activity Tracker	Detailed test population, execution status, findings and re-test position.

Controlled Artifact	Control Function
Integration & Issue Log	Material interface conditions, blockers, ownership, corrective actions and verification.
Risk & Dependency Register	Project exposure, dependency treatment and residual risk.
Decision Log	Material acceptance, governance and operationalisation decisions.
Updated Timeline	Schedule and dependency control.
Week 8 Readiness Assessment	Formal readiness and gate position.

Reporting principle: Testing status establishes what has been evidenced; acceptance determines whether the evidence is sufficient; readiness is determined only after testing, dependencies, decisions and downstream effects have been reconciled.